

not far behind. The standard UKF performs best for velocity estimation, and the spherical UKF performs best for ballistic coefficient estimation. The EKF is generally the worst performing of the four state estimators.

Table 14.1 Example 14.3 estimation errors for the extended Kalman filter, the standard unscented Kalman filter with $2n$ sigma points, and the spherical unscented Kalman filter with $(n + 1)$ sigma points. The standard UKF generally performs best. The spherical UKF performance and computational effort lie between those of the EKF and the standard UKF.

	Altitude	Velocity	Ballistic Coefficient Reciprocal
EKF	615	173	11.6
UKF	460	112	7.5
Simplex UKF	449	266	80.8
Spherical UKF	578	142	0.4

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14.5 SUMMARY

The unscented filter can give greatly improved estimation performance (compared with the extended Kalman filter) for nonlinear systems. In addition, the EKF requires the computation of Jacobians (partial derivative matrices), and the UKF does not use Jacobians. For systems with analytic process and measurement equations (such as Example 14.2), it is easy to compute Jacobians. But some systems are not given in analytical form and it is numerically difficult to compute Jacobians.

The UKF was first published in 1995 [Jul95] and since then has been expounded upon in many publications.² Although the UKF is a relatively recent development, it is rapidly finding applications in such areas as aircraft engine health estimation [Dew03], aircraft model estimation [Cam01], neural network training [Wan01], financial forecasting [Wan01], and motor state estimation [Aki03]. In addition, just as in the Kalman filter, the UKF can be implemented in a square root form to effectively increase numerical precision [Van01, Wan01]. Note that a filter based on polynomial approximations of nonlinear functions is presented in [Nor00], and it seems that the UKF is a special case of this filter.

There is a lot of room for development in the area of unscented filtering. A glance through this book's table of contents shows many specialized topics that have been applied to Kalman and H_∞ filtering, revealing a rich source of research topics for unscented filtering. These include UKF stability properties, constrained unscented filtering, unscented smoothing, reduced-order unscented filtering, robust unscented filtering, unscented filtering with delayed measurements, hybrid unscented/ H_∞ filtering, and others.

²It is interesting to note that the first journal publication of the UKF was submitted for publication in 1994, but did not appear in print until 2000 [Jul00]. Alternative technologies that are highly different than existing approaches tend to meet with resistance, but persistence (if accompanied by technical rigor) can break down barriers.

PROBLEMS

Written exercises

14.1 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = x^2$. What is $\bar{y}$? What is the first-order approximation to $\bar{y}$? What is the second-order approximation to $\bar{y}$?

14.2 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is $\bar{y}$? What is the first-order approximation to $\bar{y}$? What is the second-order approximation to $\bar{y}$? What is the third-order approximation to $\bar{y}$? What is the fourth-order approximation to $\bar{y}$?

14.3 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is the variance of y ? What is the first-order approximation to the variance of y ? What is the fourth-order approximation to the variance of y ?

14.4 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is $\bar{y}$? What is the unscented approximation to $\bar{y}$?

14.5 Consider the matrix

$$P = \begin{bmatrix} 1 & 3 \\ 3 & 9 \end{bmatrix}$$

Find an upper triangular matrix S (using only paper and pencil) such that $S^T S = P$. Find a lower triangular matrix S such that $S^T S = P$. (Note the difference between your solution to this problem and the solution to Problem 6.7.)

14.6 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is the variance of y ? What is unscented approximation to the variance of y ?

14.7 Show that for a system with an identity transition matrix, the UKF algorithm gives $\hat{x}_k^- = \hat{x}_{k-1}^+$.

14.8 Show that for a system with $y_k = x_k$, the UKF gain K_k is positive definite.

14.9 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is $\bar{y}$? Use the generalized unscented transformation to approximate $\bar{y}$ with $\kappa = 0$, $\kappa = 1$, and $\kappa = 2$.

14.10 Suppose the RV x is uniformly distributed on $[-1, 1]$, and $y = e^x$. What is the variance of y ? Use the generalized unscented transformation to approximate the variance of y with $\kappa = 0$, $\kappa = 1$, and $\kappa = 2$.

14.11 Consider the simplex sigma-point algorithm. Prove that $\sum_i W^{(i)} \sigma_i^{(j)} = 0$ (i.e., the weighted sample mean of the $\sigma_i^{(j)}$ vectors is zero).

14.12 Prove that the sum of the weights in the simplex sigma-point algorithm is equal to 1.

14.13 Consider the simplex sigma-point algorithm. Prove that the $\sum_i W^{(i)} x^{(i)} = \bar{x}$ (i.e., the weighted sample mean of the sigma points is equal to $\bar{x}$). (Hint: Use the results of Problems 14.11 and 14.12.)

Computer exercises

14.14 Design an unscented Kalman filter for the system described in Problem 13.21. Simulate the system and the filter for 60 s. Plot the estimation error for the four states. What is the experimental standard deviation of the estimation error for each of the four states? Based on the steady-state covariance matrix of the filter, what is the theoretical standard deviation of the estimation error for each of the four states? How does this compare with the extended Kalman filter results of Problem 13.21?

14.15 An inverted pendulum on a cart can be modeled as follows [Bay99, Che99].

$$\ddot{\theta} = \frac{mgl \sin \theta (M + m) - ml \cos \theta (u + ml\dot{\theta}^2 \sin \theta - B\dot{d})}{(J + ml^2)(M + m) - m^2 l^2 \cos^2 \theta}$$

$$\ddot{d} = \frac{u - ml\ddot{\theta} \cos \theta + ml\dot{\theta}^2 \sin \theta - B\dot{d}}{M + m}$$

The quantities in the system model are as follows:

- $\theta(0)$ = initial angle (0.1 rad)
- $d(0)$ = initial cart displacement (0 rad)
- m = pendulum mass (0.2 kg)
- M = cart mass (1 kg)
- g = acceleration due to gravity (9.81 m/s²)
- B = coefficient of friction between cart and ground [0.1 N/(m/s)]
- l = pendulum length (1 m)
- r = pendulum mass radius (0.02 m)
- u = external force applied to cart
- J = pendulum moment of inertia
- = $mr^2/2$

where we have assumed that the pendulum mass is concentrated in a cylinder at the end of the pendulum. Define the state of the system as $x = [d \ \dot{d} \ \theta \ \dot{\theta}]^T$. The horizontal displacement d is measured every 5 ms with a standard deviation of 0.1 m. The continuous-time process noise is $Q_c = \text{diag}(0, 0.0004, 0, 0.04)$. The system can be linearized (so that an EKF can be used to estimate the state) by assuming that θ is small, so $\cos \theta \approx 1$, $\sin \theta \approx \theta$, and $\dot{\theta}^2 \approx 0$. Suppose that the feedback control signal is given as $u = 40\theta$ and the initial state is perfectly known. Write an EKF and a UKF to estimate the state, where the control is assumed by the filters to be $\hat{u} = 40\hat{\theta}$. Plot the true states and estimated states for a 2 second simulation. Which filter appears to perform better?

